

SQL Queries & Outputs

Query 1: Market Segmentation & Relative Price Positioning

Business Context: Identify top 3 premium models per manufacturer and evaluate how far individual pricing deviates from the company's internal average price.

```
WITH RankedLaptops AS (  
    SELECT  
        Company,  
        TypeName,  
        Cpu,  
        Ram,  
        Price,  
        DENSE_RANK() OVER (PARTITION BY Company ORDER BY Price DESC) AS price_rank,  
        ROUND(AVG(Price) OVER (PARTITION BY Company), 2) AS company_avg_price  
    FROM laptops  
)  
SELECT  
    Company,  
    TypeName,  
    Cpu,  
    Ram,  
    ROUND(Price, 2) AS Price,  
    company_avg_price,  
    ROUND(Price - company_avg_price, 2) AS diff_from_company_avg  
FROM RankedLaptops  
WHERE price_rank <= 3  
ORDER BY Company, price_rank;
```

Query Output:

Company	Type Name	Cpu	Ram	Price (INR)	Company Avg Price	Diff from Company Avg
Acer	Gaming	Intel Core i7 7700HQ 2.8GHz	16	138,522.72	33,329.46	+105,193.26
Acer	Gaming	Intel Core	16	106,506.7	33,329.46	+73,177.2

		i7 7700HQ 2.8GHz		2		6
Acer	Notebook	Intel Core i7 7500U 2.7GHz	8	53,226.72	33,329.46	+19,897.26
Apple	Ultrabook	Intel Core i7 2.9GHz	16	152,274.24	83,340.50	+68,933.74
Apple	Ultrabook	Intel Core i7 2.7GHz	16	135,195.34	83,340.50	+51,854.84
Apple	Ultrabook	Intel Core i5 2.9GHz	512	106,506.72	83,340.50	+23,166.22

Query 2: Price Quartile Distribution & Hardware Correlation Analysis

Business Context: Group products into 4 equal pricing tiers (NTILE) to observe how hardware specifications (RAM, Weight) scale across price buckets.

```

WITH SpecsCategorized AS (
  SELECT
    Company,
    TypeName,
    Ram,
    Weight,
    Price,
    NTILE(4) OVER (ORDER BY Price) AS price_quartile
  FROM laptops
)
SELECT
  price_quartile,
  COUNT(*) AS total_laptops,
  ROUND(MIN(Price), 2) AS min_price,
  ROUND(MAX(Price), 2) AS max_price,
  ROUND(AVG(Ram), 2) AS avg_ram_gb,
  ROUND(AVG(Weight), 2) AS avg_weight_kg
FROM SpecsCategorized
GROUP BY price_quartile
ORDER BY price_quartile;

```

Query Output:

Price Quartile	Total Laptops	Min Price (INR)	Max Price (INR)	Avg RAM (GB)	Avg Weight (kg)
1 (Budget)	326	9,270.72	31,914.72	4.41	1.94
2 (Mid-Tier)	326	31,968.00	52,693.92	7.39	2.05
3 (Premium)	325	52,693.92	79,866.72	9.38	2.06
4 (High-End)	326	79,933.92	324,954.72	12.37	2.11

Query 3: Cumulative Inventory Valuation & 3-Company Moving Averages

Business Context: Perform time/sequence series financial rollup using explicit frame clauses (ROWS BETWEEN 2 PRECEDING AND CURRENT ROW) to observe cumulative inventory concentration across major manufacturers.

```
WITH CompanySummary AS (
  SELECT
    Company,
    COUNT(*) AS total_models,
    SUM(Price) AS total_inventory_val,
    AVG(Price) AS avg_price
  FROM laptops
  GROUP BY Company
)
SELECT
  Company,
  total_models,
  ROUND(total_inventory_val, 2) AS total_inventory_val,
  ROUND(SUM(total_inventory_val) OVER (ORDER BY total_inventory_val DESC), 2) AS
running_total_val,
  ROUND(AVG(avg_price) OVER (
    ORDER BY total_inventory_val DESC
```

```

    ROWS BETWEEN 2 PRECEDING AND CURRENT ROW
), 2) AS moving_avg_price_3_companies
FROM CompanySummary
ORDER BY total_inventory_val DESC
LIMIT 8;

```

Query Output:

Company	Total Models	Total Inventory Val (INR)	Running Total Val (INR)	Moving Avg Price (3-Company Window)
Dell	297	18,768,545.47	18,768,545.47	63,193.76
Lenovo	297	17,191,121.27	35,959,666.74	60,538.16
HP	274	15,588,146.12	51,547,812.85	59,322.45
Asus	158	9,295,162.73	60,842,975.59	57,867.92
MSI	54	4,974,276.21	65,817,251.80	69,279.14
Acer	103	3,439,645.44	69,256,897.24	61,447.00
Toshiba	48	3,242,354.40	72,499,251.64	64,353.30
Apple	21	1,750,150.50	74,249,402.14	61,428.06

Query 4: Dynamic CPU Family Categorization & Aggregation

Business Context: Parse unstandardized string attributes (Cpu) into key hardware families and evaluate model density alongside price ceilings for significant product categories (HAVING COUNT(*) > 5).

```

WITH ParsedCpu AS (
    SELECT
        CASE
            WHEN Cpu LIKE 'Intel Core i7%' THEN 'Intel Core i7'
            WHEN Cpu LIKE 'Intel Core i5%' THEN 'Intel Core i5'
            WHEN Cpu LIKE 'Intel Core i3%' THEN 'Intel Core i3'
            WHEN Cpu LIKE 'Intel%' THEN 'Other Intel'
            WHEN Cpu LIKE 'AMD%' THEN 'AMD Processor'

```

```

        ELSE 'Other'
    END AS Cpu_Brand,
    TypeName,
    Price
FROM laptops
)
SELECT
    Cpu_Brand,
    TypeName,
    COUNT(*) AS model_count,
    ROUND(AVG(Price), 2) AS avg_price,
    ROUND(MAX(Price), 2) AS max_price
FROM ParsedCpu
GROUP BY Cpu_Brand, TypeName
HAVING COUNT(*) > 5
ORDER BY Cpu_Brand, avg_price DESC;

```

Query Output:

CPU Brand	TypeName	Model Count	Avg Price (INR)	Max Price (INR)
AMD Processor	Gaming	10	51,332.16	82,530.72
AMD Processor	Notebook	50	25,699.27	58,554.72
Intel Core i3	Notebook	125	29,209.73	51,090.72
Intel Core i5	Gaming	22	54,307.72	82,530.72
Intel Core i5	Notebook	260	45,951.37	109,170.72
Intel Core i5	Ultrabook	91	69,967.62	141,186.72
Intel Core i7	Gaming	160	95,781.42	324,954.72
Intel Core i7	Workstation	26	121,556.70	261,018.72